

Quiz answer key and discussion

Instructor copy | Accept equivalent explanations and conditions.

1. Assignment and memory

Final values: a = 7, b = 3. The integer value 4 was copied into b. Later assignments update separate variables; b is not linked to a.

Ask: Which statement changes each box?

2. Values across frames

First call: 1, 1. Second call: 2, 1. total is global and keeps its value. Each call declares and initializes its own local count to 0 before adding 1.

Ask: What would happen if total = 0 were placed inside draw()?

3. Scope

It does not compile: score is local to setup(), so draw() cannot use that name. Declare int score; outside the functions, then assign score = 10; inside setup(). Do not redeclare a local int score there. A global int score = 10; with the local declaration removed also works.

Ask: Why does changing declaration to assignment matter?

4. AND and boundaries

100: false. 150: true. 200: true. 201: false. Both comparisons must be true. The left edge is excluded by >; the right edge is included by <=.

Ask: Which comparison fails at 100? At 201?

5. Separate decisions or one chain?

Version A prints 5: the first if makes x equal to 0, so the second independent if is true. Version B prints 0: the first branch runs, and the else-if branch is skipped.

Ask: Are the conditions checked using the original x or its current value?

6. Write a condition

room == 0 && (hasKey || hasPass). Example allowing entry: room=0, hasKey=true, hasPass=false. Example preventing entry: room=1, hasKey=false, hasPass=true. Equivalent correctly grouped expressions are acceptable.

Ask: Why must having a pass not bypass the room requirement? Also check room=0 with neither item, and with both items.